Please read the instructions below before the exam starts, and listen carefully to any instructions given by the invigilators. You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2023

EC114 2022/3 A 800
EC114 2021/2 A 850

Dictionaries are NOT permitted

UNIVERSITY OF READING

INTRODUCTORY MACROECONOMICS (EC114)

Three hours

Answer THREE questions as follows
Section A: Answer ALL questions
Section B: Answer ONE question
Section C: Answer ONE question

Answers should include brief, concise and to the point explanations.
All graphs must be clearly and accurately labelled.

Section A
Answer Question 1.

1. International Trade

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Terms of Trade

(ii) J-curve Effect

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) A country runs a current account surplus if it sells more of its assets abroad than it buys abroad.

(15 marks total)

(b) Consider the following macroeconomic model

Goods market:

$C = \bar{C} + mpc(Y_D)$

$I = \bar{I} - br$

$G = \bar{G}$

$Budget\ Deficit = 40$

$EX = \bar{X}$

$IM = mpm(Y_D)$

Money market:

$M^D = M_0 - M_1 r$

$M^S = \bar{M}$

$P = 2$

(i) Derive the equation for PAE. What are its slope and intercept? Derive the equation for the IS curve. What are its slope and intercept? Graph each equation.

(ii) Derive the LM curve. What are its slope and intercept? Graphically illustrate equilibrium and solve for equilibrium output and interest rates.

(iii) If the marginal propensity to consume and the marginal propensity to import were the same, how would that change your answers to part (ii)? Briefly explain the economics behind any changes.

(35 marks total)

Section B
Answer ONE of the two questions

2. Scarcity and Choice

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Marginal Rate of Transformation

(ii) Absolute Advantage

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) If the government increases the benefits for unemployed workers, resulting in more workers being unemployed and remaining unemployed for longer periods, then the PPF for the country will shift inwards.

(15 marks total)

(b) A nation with fixed quantities of resources is able to produce any of the following combinations of carpet and carpet looms (needed to make carpets). Assume there exists a fixed number of looms to start with (and hence to make carpets).

Yards of Carpet (millions)	Carpet Looms (thousands)
0	45
12	42
24	36
36	27
48	15
60	0

(i) Draw the PPF for this country. Does the principle of "increasing opportunity cost" hold for this country? Does the idea of scarcity hold for this country? Briefly explain your answers.

Question 2 continued overleaf

Question 2 cont'd

(ii) If this country chooses to produce both goods, what will happen to the PPF over time? Briefly explain.

(iii) Suppose that a new technology is developed that allows an additional 50 percent of yards of carpet to be produced by each existing loom. Illustrate on your graph from part (i) the impact of this new technology on the PPF.

(iv) Suppose that before the new technology was developed, the country produced 15 thousand looms. After the new technology, the country produced 27 thousand looms. What is the effect of the new technology on the production of carpet (be precise). Illustrate the before and after on your PPF graph.

(35 marks total)

3. Labour and Unemployment

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Labour Force

(ii) Sticky Wages

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) If all natural-gas fired electricity generation facilities are closed in the hope of slowing climate change, this will likely lead to an increase in the natural rate of unemployment.

(15 marks total)

Question 3 continued overleaf

Question 3 cont'd

(b) Suppose the following information characterizes production in the economy (for simplicity assume there is only one producer in the economy). Production is characterized by the following Cobb-Douglas production function, $F(K, L) = 4K^{1/2}L^{1/2}$ and the price at which output can be sold is 4.

(i) Derive the firm's profit function and explain how the firm would decide how much labour to hire.

(ii) Suppose the quantity of capital the firm has is 2 units. Derive the firms labour demand function and graphically illustrate it.

(iii) Suppose that labor supply is given by $L^S = 2W$. Solved for equilibrium (be precise) in the labor market and illustrate it graphically. How do the nominal and real wages compare? What is the level of unemployment in this economy?

(iv) Suppose the government sets a minimum wage of $W = 5$. What happens to labor supply and labor demand as a result of the minimum wage? What happens to the level of unemployment in the economy? Illustrate these changes graphically.

(35 marks total)

Section C
Answer ONE of the two questions

4. Short Run Keynesian Model

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Disposable Income

(ii) Loanable Funds Market

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) If individuals started holding less currency, the LM curve would shift to the right and output would increase.

(15 marks total)

(b) Consider an economy where consumption is given by $C = 100 + (1/2)Y_D - 50r$. The investment function is $I = 800 - 100r$. Government taxes are set at 400 and there is a budget deficit of 100. The demand for money in this economy depends only on the level of income; $(M/P)^D = (1/3)Y + 150$. The nominal money supply is 1800, and the average price level is 2.

(i) What is the equation of the IS curve? What is the equation of the LM curve? What are the slopes of the two curves?

(ii) Calculate all equilibrium values for this economy (Be Precise). Illustrate equilibrium.

(iii) If the government wants to balance the budget by altering taxes, what should it do? What would happen to equilibrium output and interest rates (you need only tell me the direction of change and not the exact amount)? Briefly explain the economics behind any changes to equilibrium.

Question 4 continued overleaf

Question 4 cont'd

(iv) Derive the AD curve (Be precise). Briefly explain what impact the fiscal policy in part (iii) has on this curve.

(35 marks total)

5. Long Run Keynesian Model

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Sacrifice Ratio

(ii) Okun's Law

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) In the long run, if prices increase by 10% then the nominal wage increases by at least 10%.

(15 marks total)

(b) Consider an economy which starts off in both short and long-run equilibrium. Now suppose it is the case that as a result of world events, the economy finds itself in a recession and with rising inflation.

(i) Describe a scenario which would result in the recession and inflation using words and graphs. Make sure you explain what happens using appropriate short-run graphs (and descriptions in words) for ALL macro-related markets affected.

(ii) In the absence of any policy, explain what happens to long-run equilibrium as compared to our starting point. Briefly explain (and illustrate graphically) any transition from the short run to the long run (if these differ).

Question 5 continued overleaf

Question 5 cont'd

(iii) Suppose the government decides to act to end the recession. Describe an appropriate policy to effect this outcome. Explain what would happen in both the short and long-run as a result of this policy (using both words and graphs.) How does the long run equilibrium after the policy compare to the economy's starting point?

(35 marks total)

(End of question paper)